

Syllabus for Algebra-Based Physics II: Electricity, Magnetism, and Modern Physics (PHYS135B-01)

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Abstract

The concepts of algebra-based electromagnetism, optics, and modern physics will be presented within the context of interactive problem-solving. The course will begin with the concepts of electric charge, electrostatics, electric potential, and capacitance. Next, current and DC circuits will be covered. Magnetostatics and magnetic induction follow DC circuits, concluding with AC circuits. Finally, selected topics in electromagnetic waves, optics, and modern medicine will be presented. The course work will include interactive computational exercises, analytic textbook problems, group-designed projects, and lab-based activities.

Pre-requisites: PHYS135A

Course credits, Liberal Arts Categorization: 4 Credits, None

Regular course hours and location: Monday and Wednesday, 8:50 - 10:50, SLC 232.

Instructor contact information: Discord: 918particle, Email: jhanson2@whittier.edu, Office: SLC 212, Book online appointments: <https://fgucmvjkykylvmgqfsco.10to8.com>.

Office hours: Booking service to schedule meeting: 10to8.com as above, and indicate in-person or online.

Attendance/Absence: Students needing to reschedule midterms must notify the professor a few days in advance.

Late work policy: Late work is generally not accepted, but is left to the discretion of the instructor.

Text: College Physics, 2nd Ed. <https://openstax.org/details/books/college-physics-2e>. Open-access.

Homework: Homework exercises and content will be managed by ExpertTA. This online system costs \$35 and will provide hints, feedback, and automatically grade student work. Please click this link to register: <https://reg.theexpertta.com/USB06CA-8177E0-4CL>.

Grading: The course grade will be a weighted average of assignment scores, and the weights are listed in Tab. 1. **Grade Settings:** $\geq 60\%$, $< 70\%$ = D, $\geq 70\%$, $< 80\%$ = C, $\geq 80\%$, $< 90\%$ = B, $\geq 90\%$, $< 100\%$ = A. Pluses and minuses: 0-3% minus, 3%-6% straight, 6%-10% plus (e.g. 79% = C+, 91% = A-).

ADA Statement on Disability Services: Whittier College is committed to make learning experiences as accessible as possible. If you experience physical or academic barriers due to a disability, you are encouraged to contact Student Disability Services (SDS) to discuss options. To learn more about academic accommodations, email disabilityservices@whittier.edu, call (562) 907-4825, or go to SDS which is located on the ground floor of Wardman Library.

Academic Honesty: <https://www.whittier.edu/policies/academic/honesty>

Course Objectives:

- To solve word problems in physics and mathematics, to construct mathematical models of systems
- To match mathematical models of physical systems to experimental results
- To apply logical thinking to conceptually-posed physics problems
- To practice using scientific instrumentation, to collect and analyze data, and to communicate results
- To practice written and oral expression of scientifically technical ideas

Assignment	Weight	Date
Daily exercises	10 %	Completed during class
Homework sets and labs	30 %	Weekly on Fridays
First Midterm	20 %	March 10th, 2025 (take-home style on Units 0-3)
Second Midterm	20 %	May 5th, 2025 (take-home style on Units 4-5)
Final Project Presentation	20%	April 28th and 30th, 2025 (in class)

Table 1: Grade weighting. The final project presentation may be given via **Option A** or **Option B** (see below).

Course Outline:

1. **Unit 0:** Review of pre-requisite courses, and electrostatics.
 - (a) Unit analysis, kinematics, and Newton's Laws
 - (b) Work and energy, momentum
 - (c) Electrostatics, I - **Chapters - 18.1 - 18.5**
 - i. The Coulomb Force, and Newton's Second Law for electric charges
 - ii. The concept of an electric field
 - (d) Electrostatics, II, electric potential - **Chapters 19.1 - 19.3**
 - i. Charge and electric fields in biology
 - ii. Potential energy and charge, voltage
 - iii. Potential energy and fields, point charges
2. **Unit 1:** Capacitors, current and DC circuits
 - (a) Capacitors and capacitance - **Chapters 19.4 - 19.7**
 - i. Equipotential lines, capacitance, and capacitors
 - ii. Capacitors in series and in parallel, energy considerations
 - (b) Current and DC circuits - **Chapters 20.1 - 20.5, 20.7**
 - i. DC current and resistance, Ohm's law
 - ii. Energy and power in DC current
 - iii. AC current and waveforms
 - iv. Biological example: human nerve conduction
3. **Unit 2:** DC Circuits with resistors in series and parallel, RC circuits
 - (a) DC circuit basics - **Chapters 21.1 - 21.4, 21.6**
 - i. Resistors in series and parallel, electromotive force (EMF)
 - ii. Kirchhoff's rules
 - iii. Voltmeters and ammeters
 - iv. RC circuits
4. **Unit 3:** Magnetism I
 - (a) Magnetostatics I - **Chapters 22.1 - 22.4**
 - i. Magnets, ferromagnetic and electromagnetic
 - ii. Magnetic fields and field lines, force on moving charge
 - iii. Magnetic applications I: fusion reactors
 - (b) Magnetostatics II - **Chapters 22.7 - 22.9**
 - i. Force on current carrying conductor, torque on current loop
 - ii. Ampère's Law: magnetic fields created by current
 - iii. Magnetic applications II: mass spectrometry
5. **First Midterm** - due March 10th, 2025.
 - (a) The midterm will be posted on Moodle on Friday, March 7th.
 - (b) The exam will be completed at home, with an open-book/open-note policy. The format includes multiple choice, written exercises, and design problems.
 - (c) The exam is due in PDF form on Monday, March 10th, to be submitted via Moodle.
6. **Unit 4:** Magnetism II
 - (a) Magnetic induction - **Chapters 23.1 - 23.5, 23.7, 23.9**
 - i. Induced EMF, magnetic flux
 - ii. Faraday's Law
 - iii. Motional EMF and generators, transformers
 - (b) AC circuits - **Chapters 23.10 - 23.12**
 - i. RL circuits
 - ii. RLC circuits
7. **Unit 5:** Waves, Optics, Medical Physics
 - (a) Electromagnetic waves - **Chapters 24.1 - 24.4**
 - i. Maxwell's Equations
 - ii. Electromagnetic wave production
 - iii. Electromagnetic spectrum and energy
 - (b) Optics - **Chapters 25.1 - 25.3, 25.6, 27.1 - 27.2**
 - i. Ray-tracing, reflection, refraction
 - ii. Wave interference and diffraction
 - (c) Nuclear physics in medicine - **32.1 - 32.4**
 - i. Diagnostics and medical imaging
 - ii. Biological effects of ionizing radiation
 - iii. Therapeutic uses of ionizing radiation
 - iv. Food irradiation
8. **Second midterm** - due May 5th, 2025.
 - (a) The midterm will be posted on Moodle on Friday, May 2nd.
 - (b) The exam will be completed at home, with an open-book/open-note policy. The format includes multiple choice, written exercises, and design problems.
 - (c) The exam is due in PDF form on Monday, May 5th, to be submitted via Moodle.
9. **Final project presentations**
 - (a) Given on April 28th and 30th, teams of 1-3
 - (b) Project is a self-designed physics experiment
 - (c) **Option A:** A 10-15 minute traditional presentation with several minutes for questions.
 - (d) **Option B:** A video in digital storytelling format using WeVideo, also 10-15 minutes long.



Figure 1: (Top left) Benjamin Franklin, (Top right) Father José Antonio Alzate y Ramírez. (Bottom left) Jean-Baptiste Chappé d'Aueroche, (Bottom right) Joaquín Velázquez de León.



From left to right: Carlos Román, Xavier López, and Francisco Salvatierra.

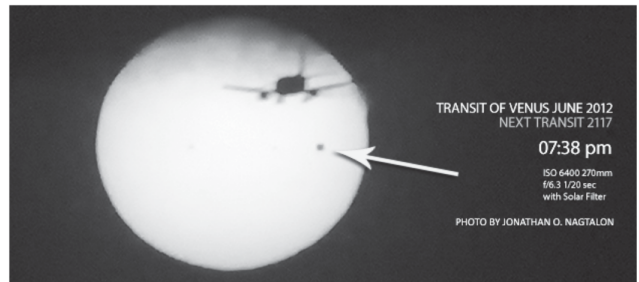


Figure 2: (Top) Carlos Román, Xavier López, and Francisco Salvatierra. (Bottom) The 2012 Venus transit, from Baja California.

- (e) Students will receive training and access for the WeVideo tools, which Whittier College provides for free.
- (f) Presentations will be graded according to the following breakdown:
 - i. For presenting results emphasizing *precision and attention to detail*, with *experimentally verifiable* content, 30%.
 - ii. For *clearly communicating* the content, including legibility and sensible graphics, 30%.
 - iii. For *meeting but not exceeding* the time-requirement with appropriate amount of content, 30%.
 - iv. For *articulating the idea with a unique style*, 10%.